

Enhanced KET-RAG: A Cost-Efficient Multi-Granular Indexing Framework for Graph-RAG

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Abstract

KET-RAG has proven itself to not only lower the indexing cost but also achieve comparable retrieval quality. However, the approach has some limitations. Using a relatively simple function to measure proximity, KET-RAG has constructed a suboptimal KNN graph in the indexing phase, which consequently limits the performance the framework could have possibly achieved. In this paper, we propose the use of the BM25 ranking function as the solution to this problem, which is able to capture the relevance between documents more precisely. The experiments conducted with two datasets, HotPotQA and MusiQue, have shown the efficiency of this modification, resulting in a significant rise in all of the metrics.

Keywords: KET-RAG, Indexing, BM25, Retrieve

1 Introduction

Within the advancements on Graph-based Retrieval Augmented Generation (GraphRAG) frameworks [1], Microsoft GraphRAG (MS GraphRAG) has shown noticeable performance compared to other GraphRAG solutions by leveraging LLMs to construct a knowledge graph (KG). However, MS GraphRAG comes with huge indexing costs, being up to 33K dollars for a 5GB real-world legal case. To address

this problem, Huang et al. introduced KET-RAG framework [2], which not only significantly reduces indexing costs, but also achieves comparable or even better retrieval quality compared to MS GraphRAG, the state-of-the-art KG RAG solution.

In order to lower indexing costs while improving retrieval and generation performance, KET-RAG does not fully materialize the knowledge graph from the text corpus like MS GraphRAG. Instead, it first selects core chunks from an immediate KNN Graph, then constructs a knowledge graph skeleton (KG Skeleton) using MS GraphRAG framework based on the selected chunks. To avoid solely depending on KG Skeleton, which could lead to information loss, KET-RAG builds a text-keyword bipartite graph by connecting every keyword to text chunks containing it. Despite KET-RAG’s computational effectiveness, its core component, the KNN graph has a quite simple building strategy, which relies on a lexical similarity function based on co-occurrence keywords. This approach might fail to capture the structural importance of keywords, thus making the graph containing noisy or less informative edges, which can negatively affect the retrieval and generation process.

In our research to improve KET-RAG framework, we focus on KNN constructing method, which is the building block of the framework. Based on what we have learned so far, we set out to develop a more reliable approach for measuring lexical similarity by leveraging the BM25 ranking function. Rather than utilizing keyword co-occurrence to measure lexical similarity, we introduced BM25-based similarity scores, which allow us to capture the relationships between text chunks with greater nuance and accuracy. Importantly, our modification preserves the original framework’s cost-efficient philosophy while enhancing generation quality.

2 Related Works

Beyond counting overlapping keywords to measure lexical similarity, we review other approaches that also compute the lexical relevance between two text chunks, in an effort to enhance the quality of the KNN graph, which significantly affects the construction of the knowledge graph skeleton.

KET-RAG builds a multi-layer index component consisting of a knowledge-graph skeleton, which is formed from core chunks selected from a KNN graph and a text-keyword bipartite graph. When constructing the KNN graph, KET-RAG combines semantic similarity with lexical similarity based on keyword co-occurrence. However, the co-occurrence count does not consider keyword importance, as it treats every text chunk with the same importance, ignores differences in text frequency, and document length. These issues can ultimately result in a KNN graph with noisy edges, negatively affecting core-chunk selection and overall retrieval.

Previous Graph-RAG systems relied heavily on lexical overlap to build KG. Li et al. [3] have combined the structural and keyword-based lexical similarity in phrase association. Wang et al. [4] have enhanced this approach with TF-IDF keyword matching [5] and TAGME entities. Although TF-IDF is more sophisticated than co-occurrence alone, there are still issues with it, such as its negligence of document length and the weight of terms.

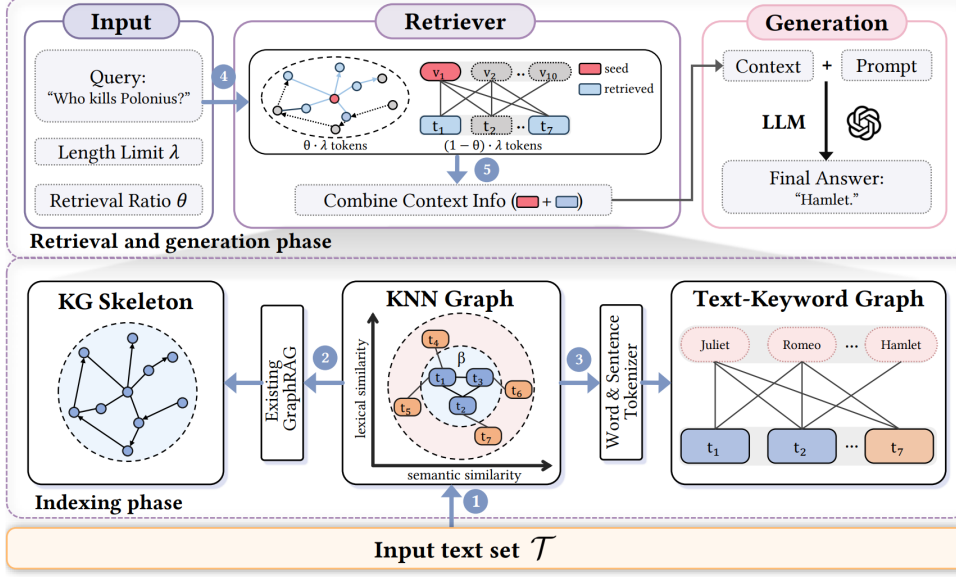


Figure 1: KET-RAG’s components and workflow [2]

The BM25 [6] ranking algorithm has been identified as one of the most effective methods for sparse term retrieval. The original BM25 ranking technique has been employed for assessing the degree of similarity between the query keywords and the documents to produce effective ranking results. According to the research work Query-Based Retrieval of German Regulatory Documents for Internal Auditing Purposes [7], in the query-based approach, the BM25 ranking algorithm clearly surpasses the TF-IDF ranking algorithm as well as the keyword search approach for the query-based search for relevant documents.

Although BM25 performs well when retrieving documents that are relevant to a query, it is never used when building knowledge graphs in current Graph-RAG systems. Existing approaches, such as KNNG-RAG, KET-RAG, and other variants instead focus on other signals to calculate similarity between documents, which are keyword co-occurrence, TF-IDF scores, or keywords extracted from LLM prompts.

3 Background

KET-RAG introduces a cost-efficient, multi-granular indexing framework for Graph-RAG, which lowers the indexing costs while preserving the retrieval and generation quality. As can be seen in Figure 1, instead of constructing a comprehensive knowledge graph from all text chunks, KET-RAG builds a knowledge graph skeleton (KG Skeleton) from a set of important chunks and a Text-Keyword Bipartite graph to ensure information sufficiency during retrieval.

The knowledge graph skeleton serves as a foundational component by applying MS GraphRAG principles. KET-RAG first constructs an intermediate K-Nearest-Neighbor (KNN) graph from the text corpus. It then identifies key text chunks having highest PageRank scores. Using the selected core chunks, KET-RAG use MS GraphRAG framework to build the KG Skeleton.

To mitigate potential information loss from solely relying on the skeleton, KET-RAG integrates a Text-Keyword Bipartite graph (Keyword-RAG) as a lightweight alternative to KG-RAG. This graph connects keywords to the text chunks in which they occur. In this framework, keywords and their corresponding text chunks function as potential entities and relationships within the knowledge graph. Keyword-RAG replicates the retrieval process of KG-RAG while significantly reducing indexing costs.

During retrieval, KET-RAG employs a local search strategy consistent with established methods. Distinct from prior approaches, it extracts ego networks from both the entity and keyword channels to improve LLM-based generation. The information retrieved from these channels is combined to form the final input for generation, constrained by a predefined context length limit. To balance the contributions of each channel, KET-RAG utilizes a retrieval ratio, denoted as θ (ranging from 0 to 1), which specifies the proportion of context sourced from Keyword-RAG and Skeleton-RAG.

4 The Proposed Method

4.1 Rationale and Observation

The knowledge graph skeleton depends heavily on the selection of core text chunks, which are picked up from an immediate KNN Graph. To construct the KNN Graph, each text chunk node is connected to $K/2$ nearest neighbors based on semantic similarity and top $K/2$ nearest nodes based on lexical similarity. Specifically, the semantic similarity is calculated as the cosine similarity between the embeddings of 2 text chunks. Meanwhile, the lexical similarity is measured by counting the number of shared keywords in 2 text chunks. Subsequently, the PageRank algorithm is applied to the constructed graph to select the most important text chunks. The core chunks are then used to construct the knowledge graph skeleton using MS GraphRAG framework.

However, the KNN Graph construction strategy has a limitation, which is the lexical similarity measure. This similarity is assessed using raw common keywords count, which treats all shared keywords equally, regardless their importance within the text chunk and the document. For example, two chunks sharing a rare and informative term might be considered irrelevant because they share a small number of common keywords. In contrast, two chunks might be consider lexically similar because they contain many identical but unimportant keyword. As a result, the constructed graph might contain noisy edges, leading to a selection of core chunks that doesn't accurately reflect the true relationships between the text chunks.

$$\begin{aligned}
& \text{② } \text{Score}_{BM25}(D, Q) = \sum_{i \in Q} IDF(q_i) * \frac{TF(D, q_i) * (k_1 + 1)}{k_1 * (1 - b + b * \frac{L_d}{L_{ave}}) + TF(D, q_i)} \\
& \text{Annotations:} \\
& \quad D = \text{Document} \\
& \quad Q = \text{Search query} \\
& \quad \text{Frequency of search term } q \text{ in document } D \rightarrow TF(D, q_i) \\
& \quad k_1 = \text{Free parameter to set the saturation of } TF(D, q_i) \\
& \quad \text{ith term of query } Q \rightarrow q_i \\
& \quad b = \text{Free parameter to set the influence of the document length} \\
& \quad L_d = \text{Length of document } D \text{ in tokens} \\
& \quad L_{ave} = \text{number of tokens of all documents}
\end{aligned}$$

Figure 2: Lexical search using BM25

4.2 Proposed Improvement

To address the above limitation, we proposed another measure of lexical similarity when constructing the KNN Graph. Instead of counting the overlapping keywords between two text chunks, we use the BM25 ranking function to compute the relevance between text chunks. This approach not only captures common keywords, but also takes into account keyword importance within the text chunk and document by incorporating term frequency, inverse document frequency, and document length normalization into the computation of the relevance between two text chunks.

Generally, BM25 is used by search engines to estimate the relevance of documents to a search query. A common use case of the BM25 function can be seen in Figure 2, displaying the application of BM25 function in lexical search. To adapt the BM25 function to our solution, we treat all text chunks in the corpus as the pseudo query, then compute its BM25 score against remaining chunks. Top $K/2$ chunks having highest score will be connected to the regarding chunk. All other components of KET-RAG, including the semantic similarity graph, the knowledge skeleton graph, and the text-keyword bipartite graph, remain unchanged.

5 Experiments

To demonstrate the effectiveness of the proposed method, the experiment answer the following research questions:

- *RQ1*: What is the result of the propose method with different datasets?
- *RQ2*: Which component of KET-RAG with BM25 takes the most time?

All experiments are conducted on a Linux machine with Ryzen 9 6900HS@3.3GHz CPU and 32GB RAM. We use the OpenAI API to access LLMs.

5.1 Dataset and Metrics

Datasets. We use two widely adopted benchmarking datasets for multi-hop QA tasks, MuSiQue [8] and HotpotQA [9]. These datasets consist of QA pairs, each accompanied by multiple paragraphs as potential relevant context. Specifically, each QA pair is associated with 2 golden paragraphs out of 20 candidate paragraphs in MuSiQue, 2 golden paragraphs and 8 distracting paragraphs in HotpotQA. We sample 500 QA instances from each dataset, with HotpotQA dataset, we only sample the medium level

Table 1: Performance comparison of KET-RAG base and KET-RAG BM25

Method	MuSiQue			HotpotQA		
	EM	F1	BERTScore	EM	F1	BERTScore
Skeleton-RAG base	8.40	11.81	42.70	—	—	—
Skeleton-RAG BM25	10.20	13.21	42.51	—	—	—
KET-RAG base	10.80	16.59	44.88	55.80	62.90	63.98
KET-RAG BM25	13.20	17.93	45.23	58.80	67.43	66.05

QA pairs. The corresponding paragraphs for all sampled instances are preprocessed into T and compiled as the external corpus for RAG [10]. The number of preprocessed paragraphs is 6,761 for MuSiQue, 20,150 for HotpotQA, resulting in an overall token count of 751,784 and 618,325 for HotpotQA.

Metrics. To evaluate answer quality generated by using pre-defined LLM based solely on the retrieved context, and to compare these answers against ground truths answer using these following standard metrics: Exact Match (EM), which measures the percentage of exact matches with the ground truth; F1score, which captures partial correctness via word-level overlap; and BERTScore, which computes semantic similarity using BERT-based embeddings.

5.2 Performance Evaluation(RQ1)

Table 1 is the result of the evaluation of KET-RAG with BM25 and without BM25. The experiment was conducted with 2 datasets MuSiQue and HotpotQA. With MuSiQue, we evaluate with 2 solutions: Skeleton-RAG, which retain only Text-keyword bipartite graph and KET-RAG, which using all two graphs in the original KET-RAG framework. Overall, the result of the all KET-RAG solutions has witnessed a significant rise in the use of BM25, which proves that it can enhance the ability of understanding the meaning of the questions and the context relevant to these questions. On MuSiQue, incorporating BM25 consistently improves EM and F1 for both methods. Skeleton-RAG increases from 8.40 to 10.20 in EM and from 11.81 to 13.21 in F1, while KET-RAG shows larger gains from 10.80 to 13.20 in EM and from 16.59 to 17.93 in F1. Regarding BERTScore, a slightly decrease is observed only for Skeleton-RAG (from 42.70 to 42.51), whereas KET-RAG benefits from a modest improvement (from 44.88 to 45.23), indicating that the BM25 can improve the overall performance of KET-RAG framework. This observation is further reinforced by the results on the HotpotQA dataset, where the benefits of integrating BM25 become even more evident. Specifically, EM score increases from 55.80 to 58.80, F1 shows a substantial improvement from 62.90 to 67.43 and BERTScore also gain a significant improvement from 63.98 to 66.05.

5.3 Runtime of Each Component(RQ2)

We analyze the time cost of different components of the KET-RAG framework based on experiments conducted on the HotpotQA dataset. The overall pipeline can be divided into two main phases: indexing phase, retrieval and generation phase. These

phases differ fundamentally in their computational characteristics, as indexing phase depends on the size of the input document collection, while retrieval and generation phase scale with the number of input questions.

At the indexing stage, KET-RAG builds two kinds of graph structures. On the one hand, constructing the KnowledgeGraph Skeleton (including KNN graph) takes 1297.41 seconds, which is the most time-consuming part in the pipeline. This cost is anticipated, since the construction of the skeleton graph consist of computing pairwise similarities and using LLMs to extract all entities and relationships across entire text corpus. In contrast, the creation of the Text–Keyword bipartite graph is much faster, taking 208.15 seconds.

The retrieval and generation phase is tested on 500 questions in the HotpotQA dataset. The retrieval stage costs 0.3215 seconds per query on average, corresponding to around two and a half minutes for all the questions. Meanwhile, answer generation is more time-consuming, with an average processing time of 3.15 seconds per question (total generation time of 26 minutes and 17 seconds). This finding suggests that generation is predominant in the online inference cost whereas retrieval is still lightweight.

6 Conclusion

In this work, we propose an enhanced method for KET-RAG, a cost-efficient multi-granular indexing framework for Graph-RAG systems. By using BM25-based similarity instead of keyword co-occurrence to construct KNN graph, our approach can measure lexical similarity between text-chunk more accurately. This leads to the result of work surpassing KET-RAG in generation quality by up to 4.53% while preserving the original framework’s cost-efficient philosophy. For future work, we plan to research the impact of the retrieval ratio parameter on generation performance, and examine adaptive strategies for dynamically assessing this parameter according to the characteristics of the queries, enabling more flexible and query-aware routing between Keyword graph and Skeleton graph.

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